

1. Explain Threat Hunting

Threat hunting is a proactive cybersecurity process in which security analysts actively search for hidden threats that may have bypassed traditional security tools such as antivirus software, firewalls, or intrusion detection systems. Instead of waiting for security alerts, threat hunters investigate networks, endpoints, and logs to identify suspicious activities before they become serious security incidents.

Threat hunting combines threat intelligence, log analysis, endpoint monitoring, and knowledge of attacker techniques to uncover advanced threats. It helps organizations detect attackers early, minimize damage, and improve their overall security posture.

Objectives of Threat Hunting

- Detect hidden cyber threats.
- Identify compromised systems.
- Reduce attacker dwell time within the network.
- Prevent data theft and ransomware attacks.
- Improve incident response and security monitoring.

Common Threat Hunting Tools

- SIEM (Security Information and Event Management)
- Microsoft Sentinel
- Splunk
- Microsoft Defender for Endpoint
- CrowdStrike Falcon
- Wireshark
- Sysmon
- MITRE ATT&CK Framework

2. Analyze an Attacker Persistence Scenario

Scenario

A company's Security Operations Center (SOC) notices that one employee's computer continues to communicate with an unknown external IP address every morning at exactly 8:00 AM. Antivirus scans do not detect any malware, but the suspicious traffic continues even after the computer is restarted.

The SOC analyst begins a threat hunting investigation by reviewing Windows Event Logs, scheduled tasks, startup programs, and registry entries. During the investigation, the analyst discovers that a malicious scheduled task was secretly created after the employee opened a phishing email attachment several weeks earlier.

The scheduled task automatically launches a hidden PowerShell script whenever the computer starts. The script connects to the attacker's command-and-control (C2) server, allowing the attacker to maintain persistent access to the infected system.

Analysis

This scenario demonstrates **attacker persistence**, where the attacker establishes a method to remain on the victim's system even after reboots. Instead of infecting the computer repeatedly, the attacker uses a scheduled task to execute malware automatically.

During the threat hunt, the analyst identifies abnormal scheduled tasks, unusual PowerShell execution, and repeated network connections to an unknown IP address. These indicators confirm that the attacker has established persistence on the endpoint.

Indicators of Compromise (IOCs)

- Unexpected scheduled tasks.
- Unknown PowerShell scripts running automatically.
- Repeated connections to unfamiliar external IP addresses.
- Suspicious registry startup entries.
- Unusual login or system activity outside normal working hours.
- Antivirus detects no malware despite suspicious behavior.

Response Steps

1. Isolate the affected computer from the network.
2. Terminate the malicious PowerShell process.
3. Remove the malicious scheduled task and any persistence mechanisms.
4. Scan the system using Endpoint Detection and Response (EDR) tools.
5. Reset the affected user's passwords.
6. Investigate whether lateral movement occurred.
7. Block the malicious IP address at the firewall.
8. Monitor the environment for any similar indicators.
9. Document the incident and update detection rules in the SIEM.